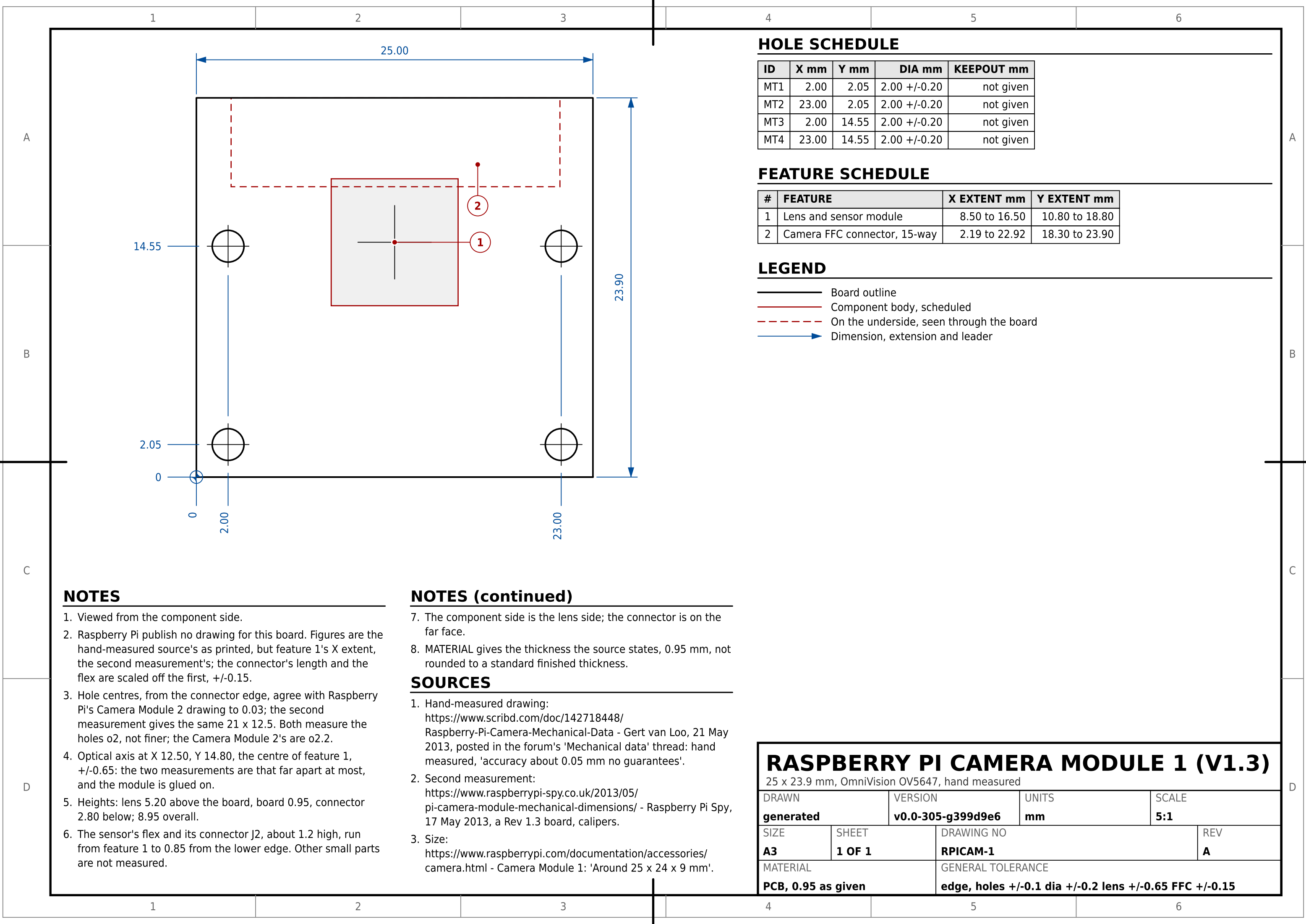




RASPBERRY PI CAMERA MODULE 1 (V1.3)

25 x 23.9 mm, OmniVision OV5647, hand measured

DRAWN generated		VERSION v0.0-305-g399d9e6	UNITS mm	SCALE 5:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO RPICAM-1			REV A
MATERIAL PCB, 0.95 as given		GENERAL TOLERANCE edge, holes +/-0.1 dia +/-0.2 lens +/-0.65 FFC +/-0.15			

1

2

3

4

5

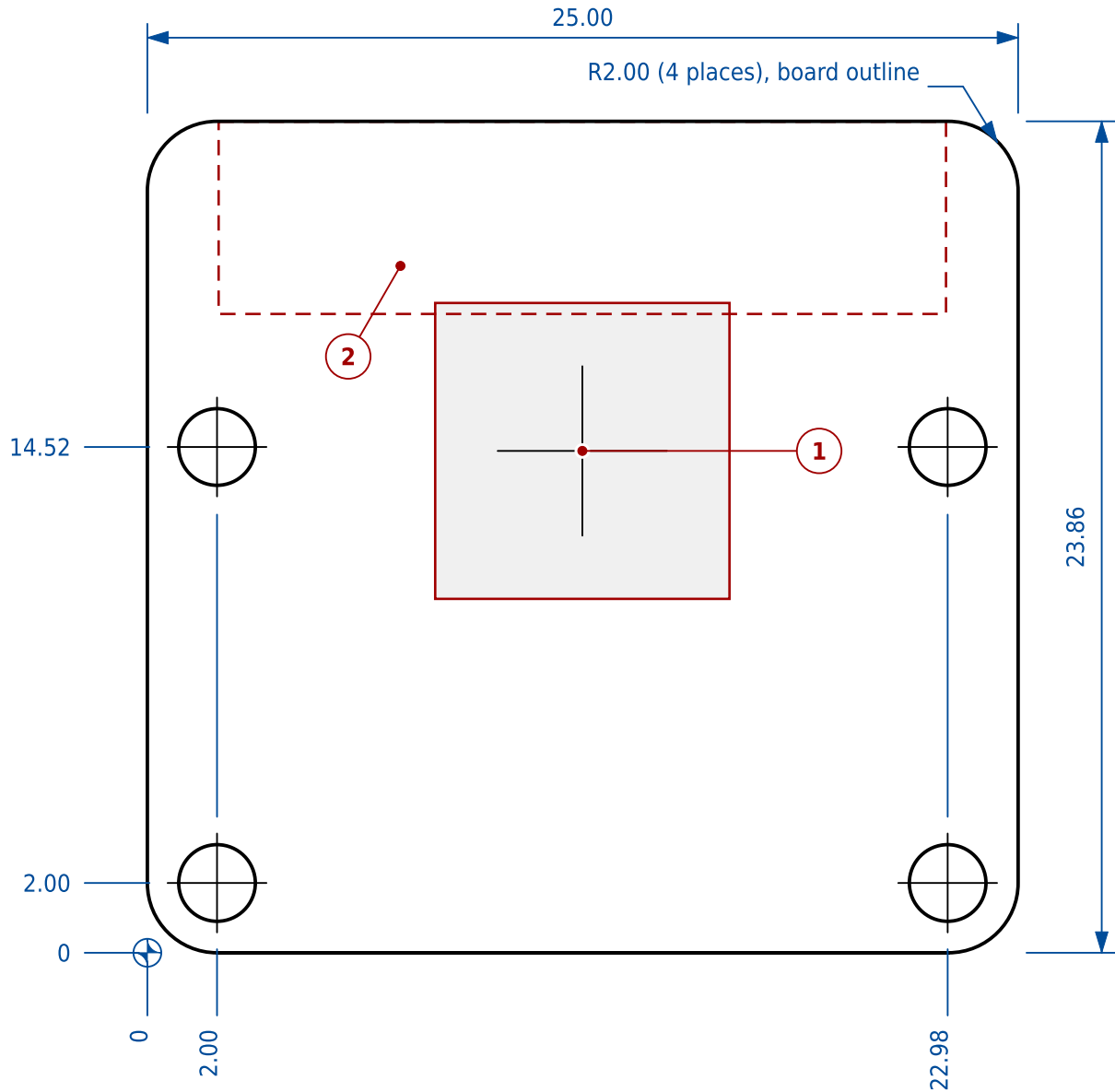
6

A

B

C

D



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	2.00	2.00	2.20	not given
MT2	22.98	2.00	2.20	not given
MT3	2.00	14.52	2.20	not given
MT4	22.98	14.52	2.20	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	Lens and sensor module	8.26 to 16.71	10.16 to 18.65
2	Camera FFC connector, 15-way	2.05 to 22.93	18.33 to 23.85

LEGEND

- Board outline
- Component body, scheduled
- On the underside, seen through the board
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are this drawing's, the same hole on every camera sheet. The two drawings are read separately and their hole patterns agree to 0.025, which is why one sheet prints 22.98 where the other prints 23.00.
- Optical axis at X 12.49, Y 14.40, the centre of feature 1. Feature 1 is dimensioned 8.5 square. Its lower body, stepped, reaches to 2.76 from the lower edge across 6.72 to 16.14.
- The source drawing is a plan view only and gives no height.
- Source read as a 1.5055:1 plot: scale from the 21.0 x 12.5 hole rectangle, overall dimensions then back within 0.031.
- Every figure printed on the source is an outlined path, not text: its geometry is machine-read, its printed dimensions are transcribed by eye.
- The component side is the lens side; the connector is on the far face.

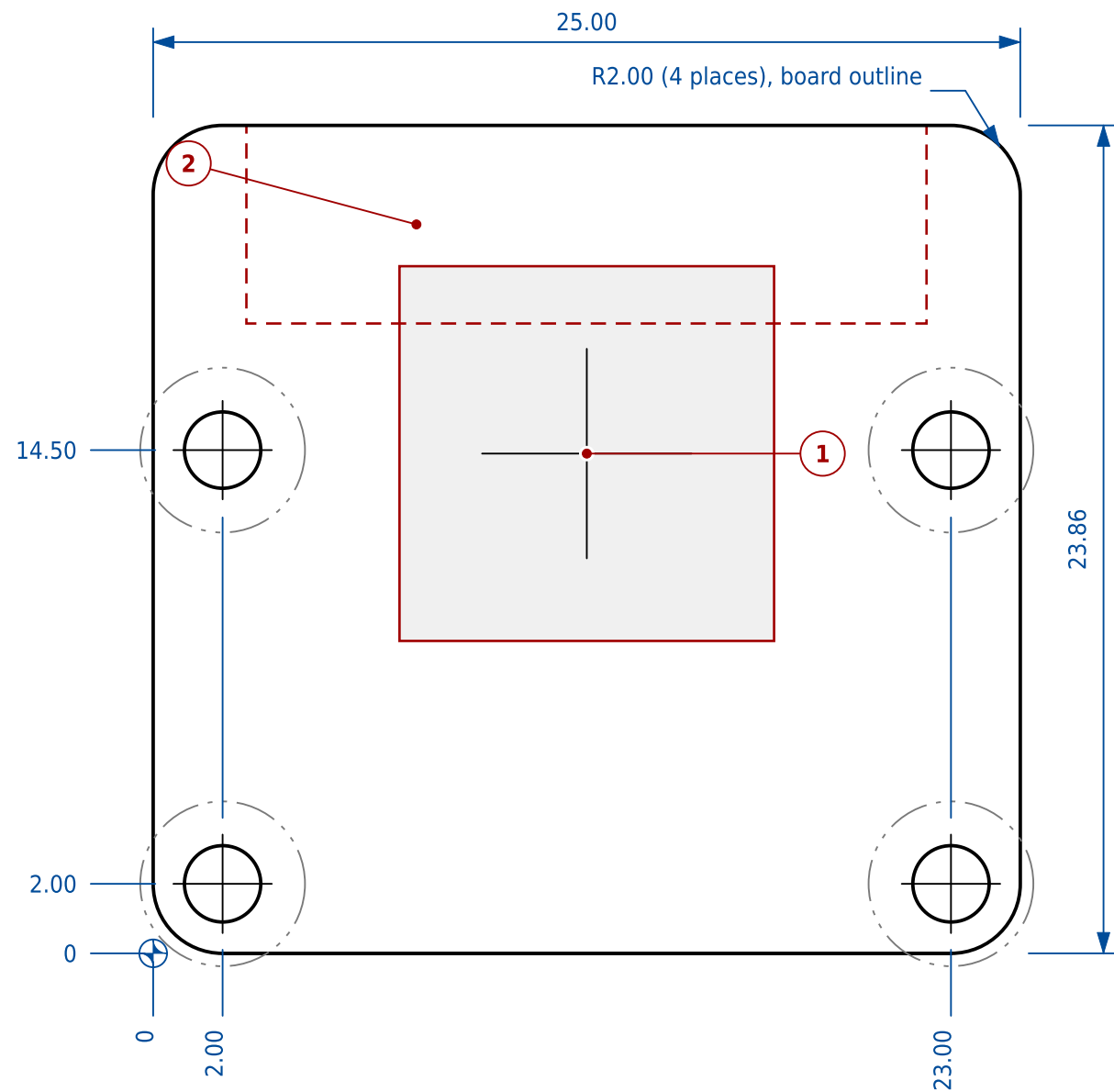
SOURCES

- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-2-mechanical-drawing.pdf> - Raspberry Pi Ltd, RPI-CAM-V2_1, 12/11/2015
- Mounting holes: <https://datasheets.raspberrypi.com/camera/camera-module-2-mechanical-drawing.pdf> - Drawing note, quoted: 4x 2.2mm diameter holes. No land or keep-out is shown.

RASPBERRY PI CAMERA MODULE 2

25 x 23.862 mm, Sony IMX219

DRAWN generated		VERSION v0.0-305-g399d9e6		UNITS mm		SCALE 5:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-2				REV A
MATERIAL PCB, thickness not stated			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	2.00	2.00	2.20	4.75
MT2	23.00	2.00	2.20	4.75
MT3	2.00	14.50	2.20	4.75
MT4	23.00	14.50	2.20	4.75

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	Lens and sensor module	7.10 to 17.90	9.00 to 19.80
2	Camera FFC connector, 15-way	2.69 to 22.30	18.15 to 23.86

LEGEND

- Board outline
- Component body, scheduled
- On the underside, seen through the board
- Hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are this drawing's, the same hole on every camera sheet. The two drawings are read separately and their hole patterns agree to 0.025, which is why one sheet prints 22.98 where the other prints 23.00.
- Optical axis at X 12.50, Y 14.40, the centre of feature 1. Clear aperture printed ø5.75 standard, ø6.95 wide; the wide drawing draws ø7.00. Feature 1's lower body, 8.9 across, reaches to 1.70 from the lower edge.
- Overall thickness printed 11.3 standard, 12 wide, lens tip to connector back; lens assembly 6.98 and 8.3 above the board. The source's own elevations scale 1.2 short, so no height is drawn here.
- Source read as a true 1:1 plot: scale from the 21.0 x 12.5 hole rectangle, overall dimensions then back within 0.004.
- The standard and wide drawings agree on the outline, holes and lens module within 0.05, so one sheet covers both.

NOTES (continued)

- The component side is the lens side; the connector is on the far face.
- MATERIAL gives the thickness the source states, 1.12 mm, not rounded to a standard finished thickness.

SOURCES

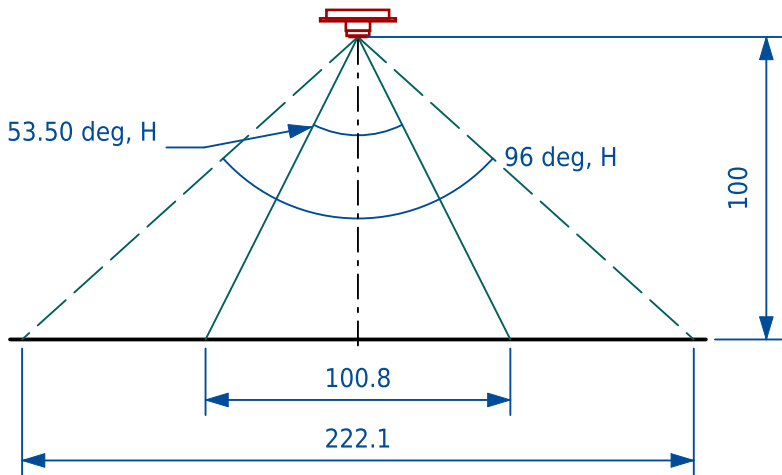
- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-3-standard-mechanical-drawing.pdf> - Raspberry Pi Ltd, RP-008153-DS-1
- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-3-wide-mechanical-drawing.pdf> - Raspberry Pi Ltd, RP-008155-DS-1
- Mounting holes: <https://datasheets.raspberrypi.com/camera/camera-module-3-standard-mechanical-drawing.pdf> - ø2.2, with an ø4.75 land round each one.

RASPBERRY PI CAMERA MODULE 3

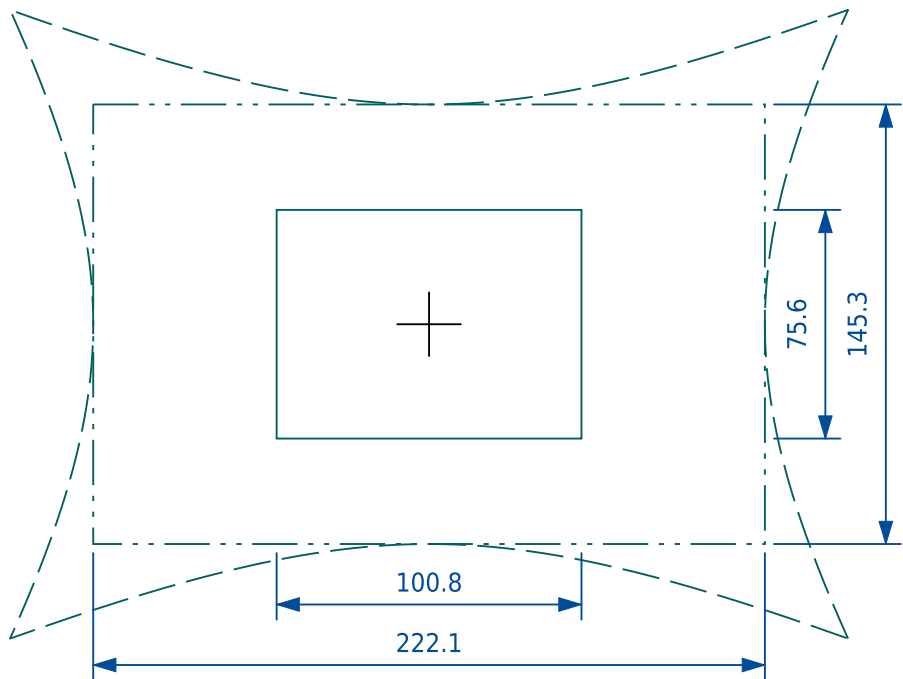
25 x 23.862 mm, standard and wide, Sony IMX708

DRAWN generated		VERSION v0.0-305-g399d9e6	UNITS mm	SCALE 5:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO RPICAM-3			REV A
MATERIAL PCB, 1.12 as given		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			

ELEVATION ACROSS THE LONG SIDE



PLAN: THE PICTURE ON A BOARD 100 BELOW



NOTES

- 65: rectilinear; its 53.50 x 41.41 come back out of 3.60 mm on the 3.6288 x 2.7216 mm active array to 0.01 deg. The 65 it is sold as is the diagonal, 64.42 on the array and 64.94 on OmniVision's image area.
- AF: 54 across implies 41.83 down, so neither declared V is exact; the lower, 41, is used. f from the 35 mm equivalent over the 43.27 mm full-frame diagonal.
- 120: the page's 120 is the DIAGONAL, so the catalogue's 120 x 90 is REJECTED: no lens sees as far across as to the corner. Its 96 x 72, for the camera without IR filter, is the diagonal x 0.8 and 0.6 like its neighbours: an equidistant split, not a measurement. YXF print D as H, H as V, V as D. The margin absorbs every row marked margin.

NOTES (continued)

- DISTORTION: a ray A/2 off the axis meets a board Z below at Z tan(A/2) whatever the lens did to it. The dashed outline is the sensor's edge walked onto the board equidistantly: its sides bow outwards, so the rectangle the angles give, which every height is worked from, is inside the picture: its corner reaches 138.6, 103.9 against the rectangle's 111.1, 72.7. At the long side's edge a pixel covers 2.23 x the board it does on the axis.
- DEPTH OF FIELD, DERIVED, thin lens, circle of confusion 2 px, 2.8 um: hyperfocal $f^2 / (N c) + f$, sharp from half of it to infinity focused there. Raspberry Pi's "Approx 1 m to infinity" is a 1.6 px circle.

SOURCES

- Raspberry Pi camera documentation; OmniVision OV5647 datasheet; Arducam 5MP OV5647 documentation; Arducam B006604 product page; Arducam B0121 product page, the B0176's predecessor; Arducam B0176 on UCTRONICS, Arducam's own store; Arducam motorized focus camera; Commonlands, OV5647 lens table; YXF YXF4Y001A1 M6 lens for OV5647 modules; Waveshare RPi Camera (G). Addresses, pinned Internet Archive captures where one exists, in raspberry_pi_camera/optics.py; every quote checked against the cached page by raspberry_pi_camera/verify_optics.py, in ASCII here: x, um, deg and infinity for the pages' signs.

FIELD OF VIEW, deg, EVERY FIGURE

	FIGURE		H	V	DIAG	
65	Raspberry Pi	DECL	53.50	41.41	--	USED
	Arducam B0033	DECL	54	41	--	margin
	3.60 mm on the active array	DERI	53.50	41.41	64.42	agrees
AF	Arducam B0176	DECL	54	44	--	margin
	Arducam B0121	DECL	54	41	--	USED
	3.67 mm, from the 35 mm equivalent	DERI	52.62	40.70	63.44	margin
120	Arducam B006604 page	DECL	--	--	120	
	Arducam catalogue, B006604	DECL	120	90	--	REJECTED
	Arducam catalogue, B006604N	DECL	96	72	--	USED
	120 diagonal, equidistant	DERI	96	72	120	agrees
	120 diagonal, rectilinear	DERI	108.36	92.20	120	margin
	120 diagonal, equisolid	DERI	94.31	69.83	120	margin
	Commonlands CIL282, 2.2 mm	DECL	96	72	122	agrees
	Commonlands CIL282, scaled to 120	DERI	94.43	70.82	120	margin
	YXF4Y001A1, 1.79 mm	DECL, relabelled	92.40	73.90	119.90	margin

FOCUS; HYPERFOCAL AND NEAR AT 2 / 1 px

	f mm	F	FOCUS	RANGE, mm	HYPER m	NEAR m
65	3.60 DECL	F2.9 DECL	Fixed	1000 to inf	1.60 / 3.20	0.80 / 1.60
AF	3.67 DERI	F2.9 ASSU	Motorized	80 to inf	1.66 / 3.32	0.83 / 1.66
120	2.17 DERI	F2.4 ASSU	Fixed	1000 to inf	0.70 / 1.40	0.35 / 0.70
WS G	3.15 DECL	F2.35 DECL	Adjustable	100 to inf	--	--

AF, FOCUSED AT Z: SHARP FROM, TO, mm

Z	NEAR	FAR	DEPTH
80	76.5	83.9	7.4
100	94.5	106.2	11.7
150	137.8	164.5	26.7

LEGEND

- The 65 deg lens: rays, and its picture on the board
- The 120 deg lens: rays, and its picture on the board
- The rectangle the 120 deg lens's angles give
- Camera Module v1.3
- Dimension, extension and leader

OV5647 LENSES AND FOCUS

Camera Module v1: stock, autofocus and 120 degree lenses

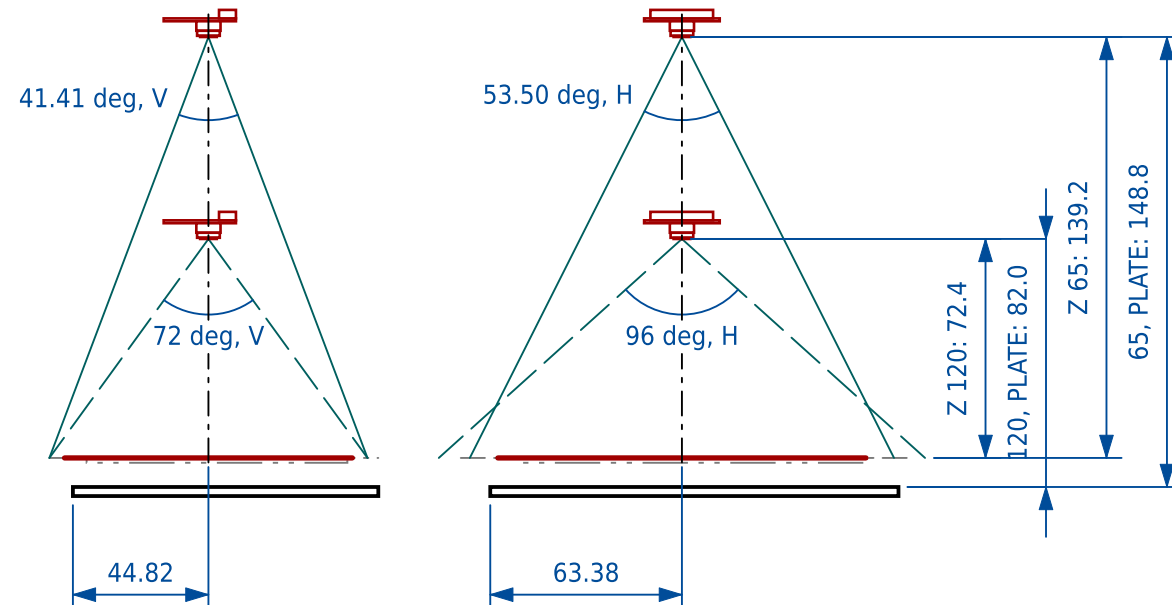
DRAWN	VERSION	UNITS	SCALE
generated	v0.0-352-g967eb38	mm	1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-LENS	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
OV5647 lenses	angles as declared, DERI and ASSU as flagged		

NOTES (continued)

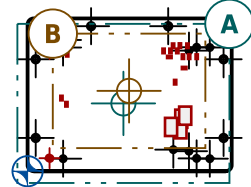
- FIXED FOCUS, ASSUMED at 2 m, the stock lens's hyperfocal distance; the 120's is 0.70 m, so its 1 m reads copied. At 100 mm a point spreads to 30 px (1.1 mm on the board) at 65 and 13 px (0.8 mm) at 120: out of focus, and within two pixels the same set at infinity.
- AF: "80mm to infinity", the B0121 "4 cm"; the 80 is used. Arducam's catalogue lists no 120 deg motorised OV5647.
- WS G, Waveshare's RPi Camera (G): 160 deg diagonal on Waveshare's page, 120 across on The Pi Hut's; its 3.15 mm gives 82.5 diagonal equidistantly, so it does not hold together and is not drawn.

END ELEVATION

FRONT ELEVATION



PLAN, 1:5



NOTES

1. First angle; the end elevation is seen from the right. The camera, a Camera Module v1.3 from its own data, drawn per lens and fitted one at a time, looks straight down, in a mode reading the whole sensor. Z is to its entrance pupil, which nobody locates; ASSUMED behind the lens face by at most the lens's 5.20 mm, so set the FACE at Z and the picture is up to 7.2% larger, never smaller.

X0 Y0

NOTES (continued)

2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round, turned whichever way needs the lower camera; LONG says which. One margin absorbs both where the stand ends up -- 5.00 mm of lateral error, or a lean of 1.8 deg at frame A's 65 Z or 2.7 at its 120, measured at the picture's edge -- and what is not known about the lens, below: what one uses the other cannot.
3. PLANE, which Z is measured from: A/B from the DEMO BOARD's top face, not the plate's -- 8 mm standoffs and a 1.56 to 1.60 mm board put it 9.56 to 9.60 above the plate face; Z is from the higher. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. HEADROOM: a part standing on a board, anywhere in its envelope stays in frame A's picture up to 13.2 mm at 65, 6.9 mm at 120 above the plane Z is set from. Higher, raise the camera.
5. FOCUS. Both fixed lenses are declared "Approx 1 m to infinity" (Raspberry Pi) and "Focus Distance 1 m to infinity" (Arducam), and every Z here is nearer: on either the board is OUT OF FOCUS. At frame A's Z a point spreads to 21 px, 1.1 mm on the board, at 65, and 19 px, 0.8 mm, at 120 (DERIVED, thin lens ASSUMED set at 2 m). The autofocus AF, Arducam's B0176, is declared "80mm to infinity": every Z of 80 or more is in focus once it has focused; at frame A's its depth of field is 129.9 to 153.4 (DERIVED, F2.9 ASSUMED). Arducam's catalogue lists no 120 deg OV5647 with a motorised lens.

NOTES (continued)

6. The 120 deg lens is a fisheye, 120 deg on the diagonal: 96 x 72 is Arducam's split of that diagonal, equidistant, not a measurement. Its barrel distortion bows the picture's edges outwards on the board, so the frame's corners are inside it; the margin absorbs the narrower lenses the evidence allows, 120 diagonal, equisolid, 94.3 x 69.8, leaving 2.9 mm; Commonlands CIL282, scaled to 120, 94.4 x 70.8, leaving 3.9 mm; YXF4Y001A1, 1.79 mm, 92.4 x 73.9, leaving 6.9 mm. See RPICAM-LENS.
7. The "65 deg" name is the stock lens's DIAGONAL, which no vendor prints: $2 \times \arctan(\sqrt{3.6736^2 + 2.7384^2}) / 2 / 3.60 = 64.94$ deg on OmniVision's image area, 64.42 on the pixel array. Every Z here is from the angle along each axis. 65 deg across frame A's long side would give Z 110.1 and lose 14.6 mm off each end.
8. Quotes are transliterated to ASCII: x, um and infinity for the signs the cached pages print.
9. Frame A, Every board, any revision: Every revision's assembled envelope -- outline, connectors, Pmod bodies -- in plate coordinates, in one: a camera fixed over the plate serves them all.
10. Frame B is the union of every LED and 7-segment over all five revision families. They are not clustered -- 90.91 x 62.79 of a 135 x 101 plate -- so frame B buys little over frame A.

SOURCES

1. Plate geometry: [tinypoint/tinypoint/mounting_plate/plate.py](https://github.com/tinypoint/tinypoint/blob/master/mounting_plate/plate.py) - Outline, placements and standoff; see TT-MP-PLATE.

FRAMES, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y
A	Every board, any revision	140.27 x 105.20	X	63.38	44.82
B	Every LED and 7-seg	100.91 x 75.69	X	67.13	53.12

Z ABOVE THE FRAME PLANE, mm, AND IN FOCUS THERE?

	Z 65	FIXED	Z AF	MOTOR	Z 120	FIXED	MOTOR
A	139.2	no	140.7	yes	72.4	no	none listed
B	100.1	no	101.2	yes	52.1	no	none listed

LENSES, deg, AS USED: SEE RPICAM-LENS

	MODULE	H	V	DIAG	f mm	F	FOCUS FROM
65	RPi Camera v1.3	53.50	41.41	64.4 DERI	3.60 DECL	F2.9 DECL	Fixed, 1 m
AF	Arducam B0176	54	41	64.6 DERI	3.67 DERI	F2.9 ASSU	Motor, 80 mm
120	Arducam B006604	96	72	120.0 DECL	2.17 DERI	F2.4 ASSU	Fixed, 1 m

LEGEND

- Subject outline
- LED, display or connector
- - - Adjacent part or connector body
- Frame A's target, on the plane Z is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- The 120 deg lens's field of view, 96 x 72, to its picture's edge
- - - Frame A, the rectangle the picture must cover
- - - Frame B, the rectangle the picture must cover
- Dimension, extension and leader

SOURCES (continued)

2. Board and indicator positions: [tinypoint/boards.py](https://github.com/tinypoint/tinypoint/blob/master/boards.py) - Outlines, connectors, Pmod bodies, LEDs and 7-segment displays, in plate coordinates.
3. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

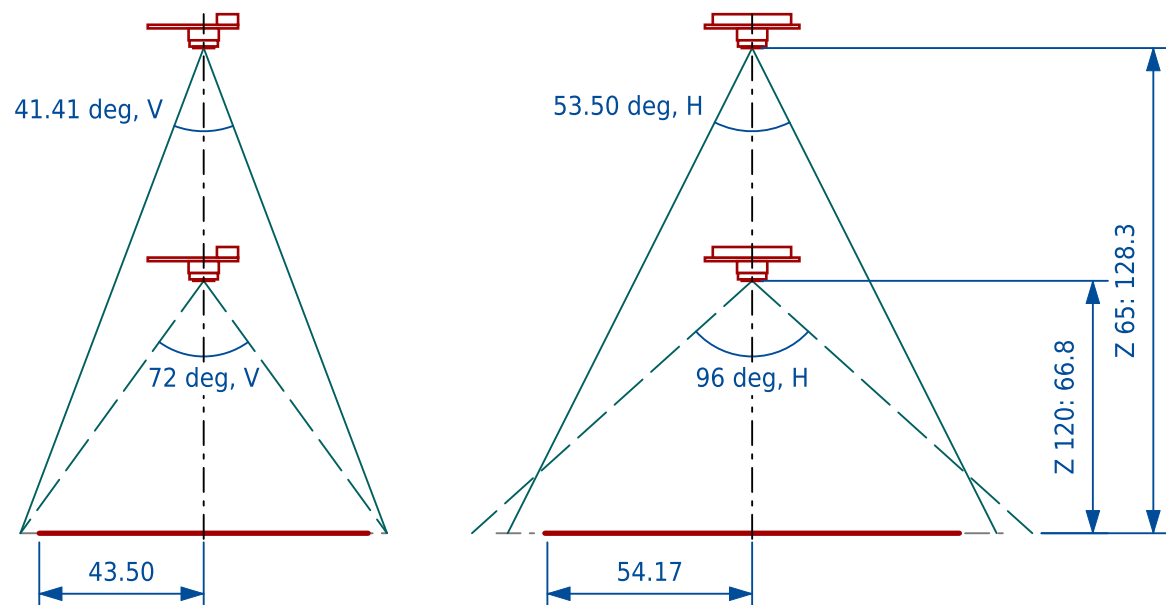
CAMERA OVER THE TT MOUNTING PLATE

Camera Module OV5647, 65 and 120 degree lenses

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-352-g967eb38	mm	1:2.5, PLAN 1:5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-OVER-PLATE	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
TT Mounting Plate	plate +/-0.20, LEDs +/-0.10, Z DERIVED		

END ELEVATION

FRONT ELEVATION



PLAN, 1:2.5

NOTES

1. First angle; the end elevation is seen from the right. The camera, a Camera Module v1.3 from its own data, drawn per lens and fitted one at a time, looks straight down, in a mode reading the whole sensor. Z is to its entrance pupil, which nobody locates; ASSUMED behind the lens face by at most the lens's 5.20 mm, so set the FACE at Z and the picture is up to 7.8% larger, never smaller.

FRAMES, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y
A	Whole board	129.33 x 97.00	X	54.17	43.50
B	LD0-LD7	32.68 x 24.51	X	19.50	6.83

Z ABOVE THE FRAME PLANE, mm, AND IN FOCUS THERE?

	Z 65	FIXED	Z AF	MOTOR	Z 120	FIXED	MOTOR
A	128.3	no	129.7	yes	66.8	no	none listed
B	32.4	no	32.8	no	16.9	no	none listed

LENSES, deg, AS USED: SEE RPICAM-LENS

	MODULE	H	V	DIAG	f mm	F	FOCUS FROM
65	RPi Camera v1.3	53.50	41.41	64.4 DERI	3.60 DECL	F2.9 DECL	Fixed, 1 m
AF	Arducam B0176	54	41	64.6 DERI	3.67 DERI	F2.9 ASSU	Motor, 80 mm
120	Arducam B006604	96	72	120.0 DECL	2.17 DERI	F2.4 ASSU	Fixed, 1 m

LEGEND

- Subject outline
- LED, display or connector
- - - Adjacent part or connector body
- Frame A's target, on the plane Z is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- The 120 deg lens's field of view, 96 x 72, to its picture's edge
- Frame A, the rectangle the picture must cover
- Frame B, the rectangle the picture must cover
- Dimension, extension and leader

NOTES (continued)

2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round, turned whichever way needs the lower camera; LONG says which. One margin absorbs both where the stand ends up -- 5.00 mm of lateral error, or a lean of 2.0 deg at frame A's 65 Z or 2.9 at its 120, measured at the picture's edge -- and what is not known about the lens, below: what one uses the other cannot.
3. PLANE, which Z is measured from: A/B from the subject's own top face. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. FOCUS. Both fixed lenses are declared "Approx 1 m to infinity" (Raspberry Pi) and "Focus Distance 1 m to infinity" (Arducam), and every Z here is nearer: on either the board is OUT OF FOCUS. At frame A's Z a point spreads to 23 px, 1.1 mm on the board, at 65, and 20 px, 0.8 mm, at 120 (DERIVED, thin lens ASSUMED set at 2 m). The autofocus AF, Arducam's B0176, is declared "80mm to infinity": every Z of 80 or more is in focus once it has focused; at frame A's its depth of field is 120.6 to 140.4 (DERIVED, F2.9 ASSUMED). Arducam's catalogue lists no 120 deg OV5647 with a motorised lens.

NOTES (continued)

5. The 120 deg lens is a fisheye, 120 deg on the diagonal: 96 x 72 is Arducam's split of that diagonal, equidistant, not a measurement. Its barrel distortion bows the picture's edges outwards on the board, so the frame's corners are inside it; the margin absorbs the narrower lenses the evidence allows, 120 diagonal, equisolid, 94.3 x 69.8, leaving 3.1 mm; Commonlands CIL282, scaled to 120, 94.4 x 70.8, leaving 4.0 mm; YXF4Y001A1, 1.79 mm, 92.4 x 73.9, leaving 6.2 mm. See RPICAM-LENS.
6. The "65 deg" name is the stock lens's DIAGONAL, which no vendor prints: $2 \times \arctan(\sqrt{3.6736^2 + 2.7384^2} / 2 / 3.60) = 64.94$ deg on OmniVision's image area, 64.42 on the pixel array. Every Z here is from the angle along each axis. 65 deg across frame A's long side would give Z 101.5 and lose 13.5 mm off each end.
7. Quotes are transliterated to ASCII: x, um and infinity for the signs the cached pages print.
8. Frame A, Whole board: The assembled envelope: the outline together with the connectors and Pmod bodies that overhang it.
9. Frame B, LD0-LD7: The four tri-colour LEDs LD0-LD3 and the four single LD4-LD7, in two rows on a 7.00 mm pitch.

NOTES (continued)

10. Which LED row is which is not named by any Digilent source; FPGA-ARTY-A7 takes the row nearest the edge as the tri-colour LD0-LD3 by package size. Either way both rows are inside the frame, so the framing does not turn on it.

SOURCES

1. Board geometry: fpga/boards.py - Outline, Pmod hosts, connectors and LED rows; see FPGA-ARTY-A7.
2. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE ARTY A7

Camera Module OV5647, 65 and 120 degree lenses

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-352-g967eb38	mm	1:2, PLAN 1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-OVER-ARTY	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
Digilent Arty A7	DXF +/-0.20, plot bodies +/-0.30, Z DERIVED		